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MTH301 Calculus II

Mid Term Examination – Spring 2006

Time Allowed: 90 Minutes

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.

2. Do not ask any questions about the contents of this examination from anyone.

a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.

b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

c. Write all steps, missing steps may lead to deduction of marks.

3. Calculator is allowed.

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****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

Question No. 1	**Marks : 5**

Find the directional derivative $D_u f(1, 1,1)$ of $f(x, y, z) = x^2 z^3 y^2 z - xyz$ in the direction of $u = -\langle 1, 0,3\rangle$ at the point $(1, 1,1)$.

Question No. 2	**Marks : 1**

$$\int \frac{1}{4x^2 + x}\, dx = \ln |4x + x| + c$$

? True.

? False

Question No. 3 **Marks : 10**

Find

$\frac{?z}{?s}$ and $\frac{?z}{?t}$, where $z(s, t) f(r(s, t), (s, t))$, $f(r,) e \sin(3)^{2r}$, $r(s, t) = -st t$, $(s, t)? = \sqrt{s^2 + t^2}$

Question No. 4 **Marks : 10**

Compute the volume of the solid bounded by the given surface

$z = -1y$, $z = 0$, $y = 0$, $x = 1$, $x = 2$

Question No. 5 **Marks : 1**

Not all critical points need to be local extrema.

? True.

? False

Question No. 6 **Marks : 1**

$f(x, y) = \sqrt{3x^2 - y^2}$ is continuous at

? $x^2+ = y^23$

? x^2y^2

? x^2y^2

? x^2y^2

Question No. 7 **Marks : 5**

Find an equation of the plane through the origin that is parallel to the plane:

$4x - 2y + 7z + 12 = 0$

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Question No. 8 **Marks : 10**

Use double integrals to compute the area of the region bounded by the curves:

$y = x^2$ and $y = 8 - x^2$

Question No. 9 **Marks : 1**

If f (is +ve then the graph of function is concave up.

? True.

? False

Question No. 10 **Marks : 1**

Direction cosines of straight line through A(1,1,1) and B(0,0,-1) are

? $\frac{2}{3}, \frac{2}{3}, \frac{1}{3}$

? $\frac{2}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

? $\frac{3}{\sqrt{1}}, \frac{3}{\sqrt{1}}, \frac{3}{\sqrt{2}}$

? $\frac{6}{6}, \frac{6}{6}, \frac{6}{6}$

? None of these.

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